PROJECT REPORT

Telecom Customer Churn Analysis Using Python

Identifying why customers leave and how to retain them

Dataset: Telecom Customer Churn (Maven Analytics)

Records / Columns: 7,043 customers / 38 columns

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

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1. Project Overview

Customer churn occurs when customers stop using a company's services. Understanding churn helps telecom companies improve customer satisfaction, retain existing customers, and reduce revenue loss.

This project analyzes telecom customer data using Python to understand customer behaviour, identify the major reasons for churn, and provide business recommendations to improve customer retention.

2. Dataset Information

Property	Details
Dataset name	Telecom Customer Churn
Source	Maven Analytics – Telecom Customer Churn (mavenanalytics.io/data-playground/telecom-customer-churn)
File format	CSV
Total records	7,043 customers
Total columns	38
Customer status	Stayed: 4,720 (67.02%) | Churned: 1,869 (26.54%) | Joined: 454 (6.45%)

The dataset includes customer demographics, service subscriptions, contract details, billing information, tenure, customer status, churn categories, and churn reasons.

3. Project Objectives

- Calculate the overall customer churn rate.
- Identify customer groups with higher churn rates.
- Analyze the relationship between customer tenure and churn.
- Understand how contract types influence customer churn.
- Analyze monthly charges and total revenue.
- Identify the most common reasons for customer churn.
- Understand how internet and additional services influence customer retention.

Provide business recommendations to reduce customer churn.

4. Tools and Technologies

Tool	Purpose
Python	Data analysis and programming
Pandas	Data cleaning and manipulation
NumPy	Numerical calculations
Matplotlib	Data visualization
Seaborn	Statistical data visualization
Google Colab	Development environment

5. Project Workflow

5.1 Data Loading

- Imported the dataset using Pandas.
- Examined the dataset structure, columns and data types.
- Checked the number of records and attributes.

5.2 Data Cleaning

- Identified and handled missing values, checked duplicates, investigated negative Monthly Charge values and replaced missing values where appropriate.

- Duplicate records: none found.
- Negative Monthly Charge values: 120 records were identified and flagged for investigation; they were retained in the data.

5.3 Data Transformation

- Converted Customer ID into string format.
- Created a separate dataset for existing customers (Stayed and Churned: 6,589 records). Joined customers were excluded because they are new and have not yet had the chance to churn.
- Created a numerical Churn Flag (1 = Churned, 0 = Stayed).

5.4 Feature Engineering

- **Tenure Group:** 0–12, 13–24, 25–48 and 49–72 months.
- **Age Group:** 18–25, 26–40, 41–60 and Above 60 years.
- **Monthly Charge Category:** Low (up to 35), Medium (35–70) and High (above 70).
- **Additional Services Count:** number of add-on services (0–7) a customer subscribes to.
- **Churn Flag:** binary indicator used to calculate churn rates.

5.5 Exploratory Data Analysis

Analysis covered the overall churn rate, customer demographics, tenure, contract types, monthly charges and total revenue, churn reasons, internet services and additional services.

5.6 Statistical Analysis

Mean, median, mode, variance, standard deviation, skewness and kurtosis were calculated for the key numerical variables (existing customers only).

Variable	Mean	Median	Mode	Variance	Std Dev	Skew	Kurt
Age	46.76	46.00	47.00	283.63	16.84	0.15	-1.02
Tenure in Months	34.50	32.00	1.00	574.50	23.97	0.16	-1.39
Monthly Charge	65.03	71.05	20.05	967.26	31.10	-0.35	-1.06
Total Revenue	3,235.22	2,376.45	24.80	8,148,741	2,854.60	0.86	-0.31
Number of Referrals	2.02	0.00	0.00	9.14	3.02	1.40	0.58

- **Age:** average of about 47 years, median 46; the distribution is fairly balanced around its average.
- **Tenure:** average 34.5 months, median 32; the most common tenure is 1 month, and tenure varies widely.
- **Monthly Charge:** the mean (65.03) is below the median (71.05), partly because of the negative values that need investigation.
- **Total Revenue:** the mean (3,235.22) is above the median (2,376.45), showing that some customers contribute considerably higher revenue.
- **Referrals:** median and mode are both zero, so most customers have not referred anyone.

Comparison by customer status:

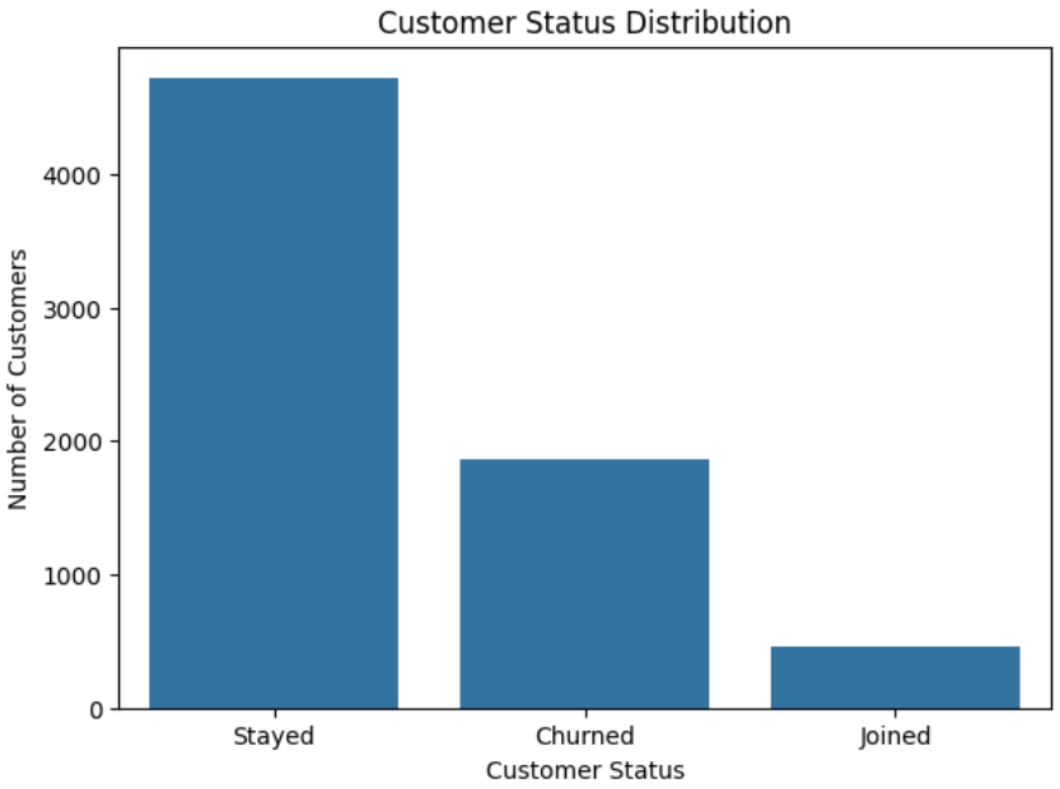
Status	Tenure mean	Tenure median	Monthly mean	Monthly median	Revenue mean	Revenue median
Churned	17.98	10.0	73.35	79.50	1,971.35	893.55
Stayed	41.04	42.0	61.74	65.60	3,735.68	2,960.11

6. Objective-Based Visualizations

This section presents nine visualizations. Each one is linked to a project objective and is followed by its Objective, Interpretation, Key Insight and Business Insight.

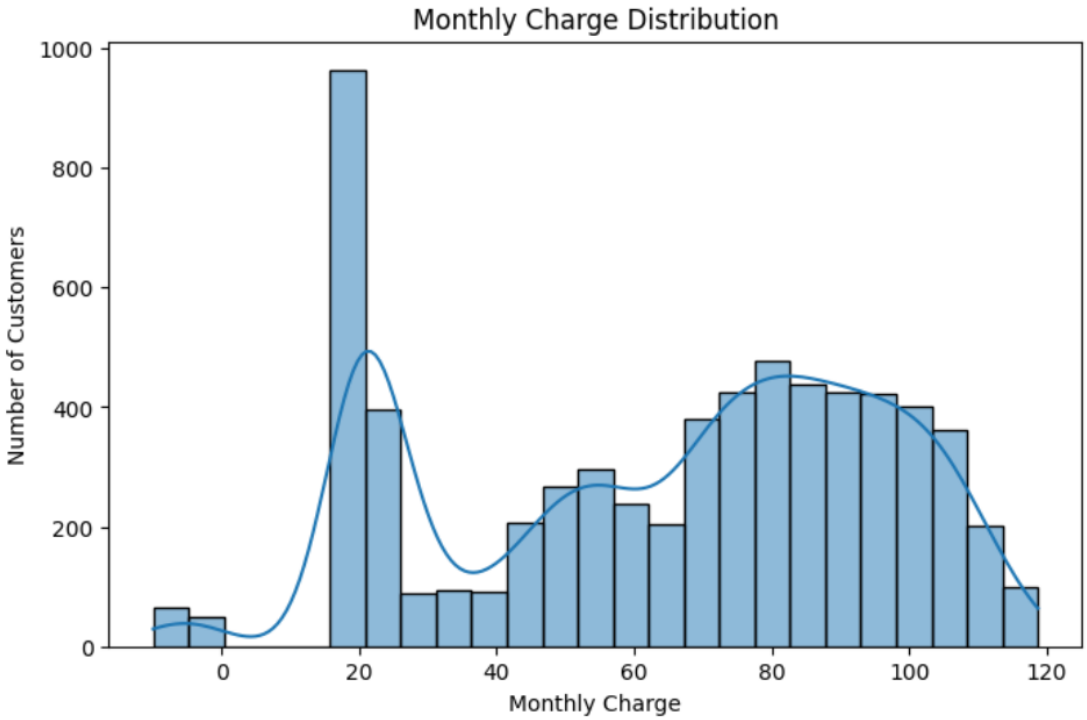
No.	Visualization	Project objective addressed
1	Customer status distribution and overall churn rate	Overall churn rate
2	Churn by gender, age group and marital status	Customer groups with higher churn
3	Churn rate by tenure group	Tenure and churn
4	Average and median tenure by customer status	Tenure and churn
5	Churn rate by contract type	Contract type and churn
6	Monthly charge and total revenue by status	Monthly charges and revenue
7	Top 10 churn reasons	Reasons for churn
8	Churn rate by internet type	Internet services and retention
9	Churn rate by number of additional services	Additional services and retention

Visualization 1: Customer Status Distribution and Overall Churn Rate



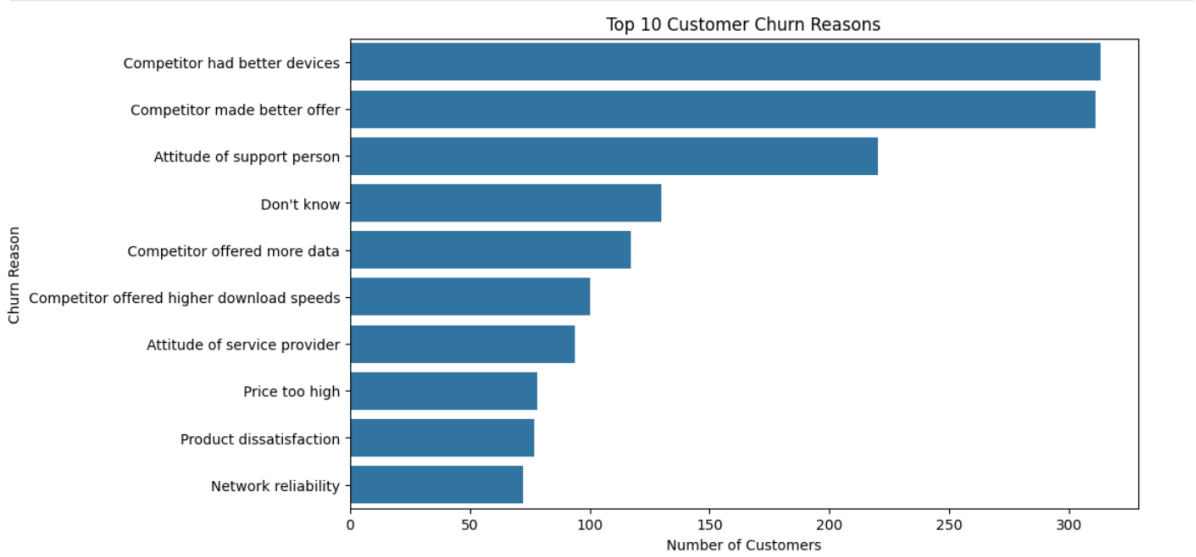
Objective	To understand the distribution of customers based on their status.
Interpretation	Customer tenure varies across different periods. Some customers have recently joined. Some customers have been using the services for several years.
Key Insight	Customer tenure helps identify the relationship between customer loyalty and churn.
Business Insight	The company should provide special offers and better support to newly joined customers to improve long-term retention.

Visualization 2: Monthly Charge Distribution



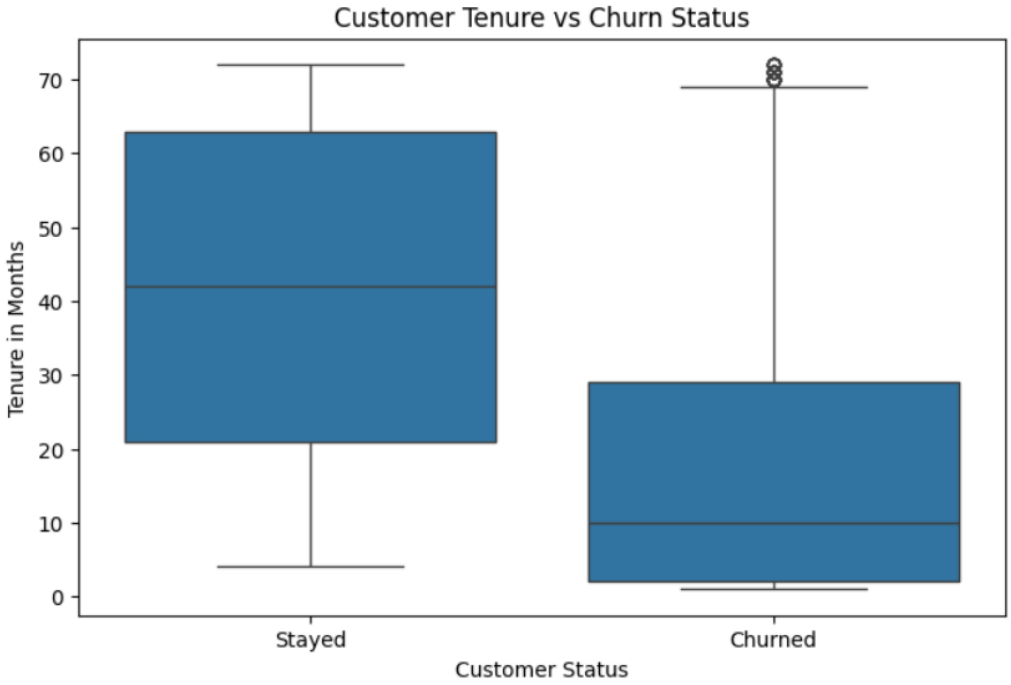
Objective	To understand customer monthly spending patterns.
Interpretation	Monthly charges vary among customers. The chart shows different customer spending levels. Negative monthly charge values are also present in the dataset.
Key Insight	Customers have different monthly spending patterns, and unusual negative values require further investigation.
Business Insight	The company should review billing accuracy and provide suitable pricing plans based on customer needs.

Visualization 3: Top 10 Customer Churn Reasons



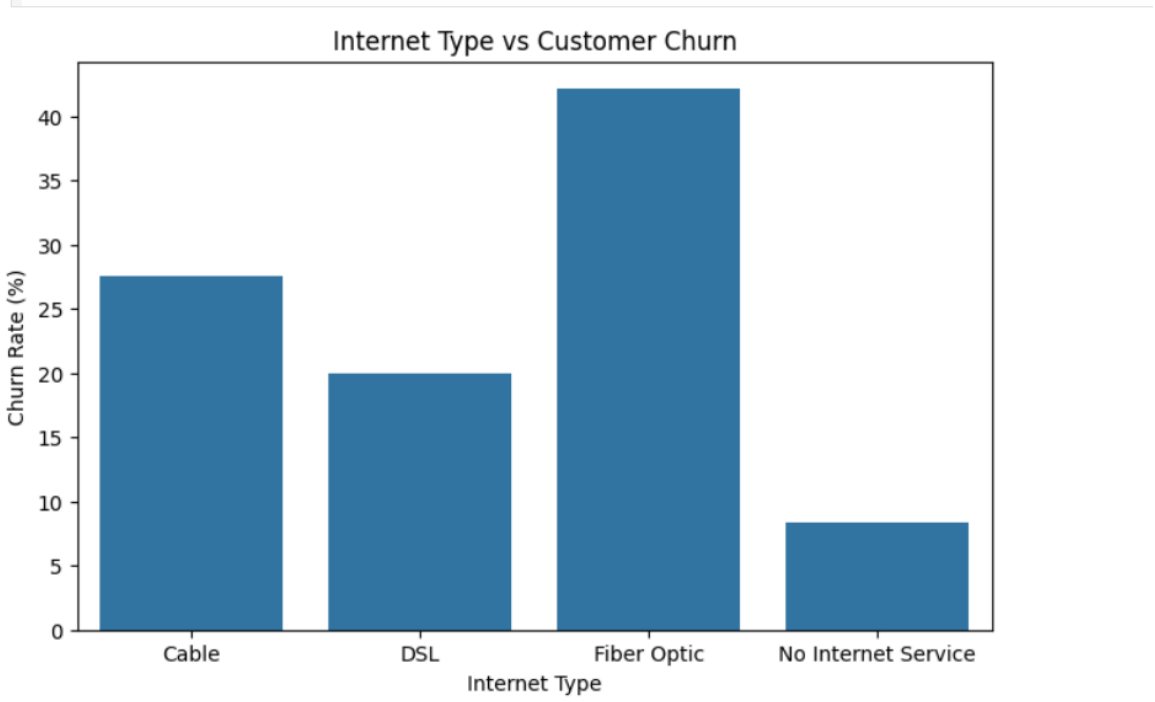
Objective	To identify the most common reasons why customers leave the company.
Interpretation	• Better devices from competitors are the top reason. • Better competitor offers are another major reason. • Customer support attitude is also an important reason.
Key Insight	Competitor-related issues are the most frequently reported reasons for customer churn.
Business Insight	The company should provide competitive offers, improve product quality and strengthen customer support services.

Visualization 4: Tenure vs Customer Churn



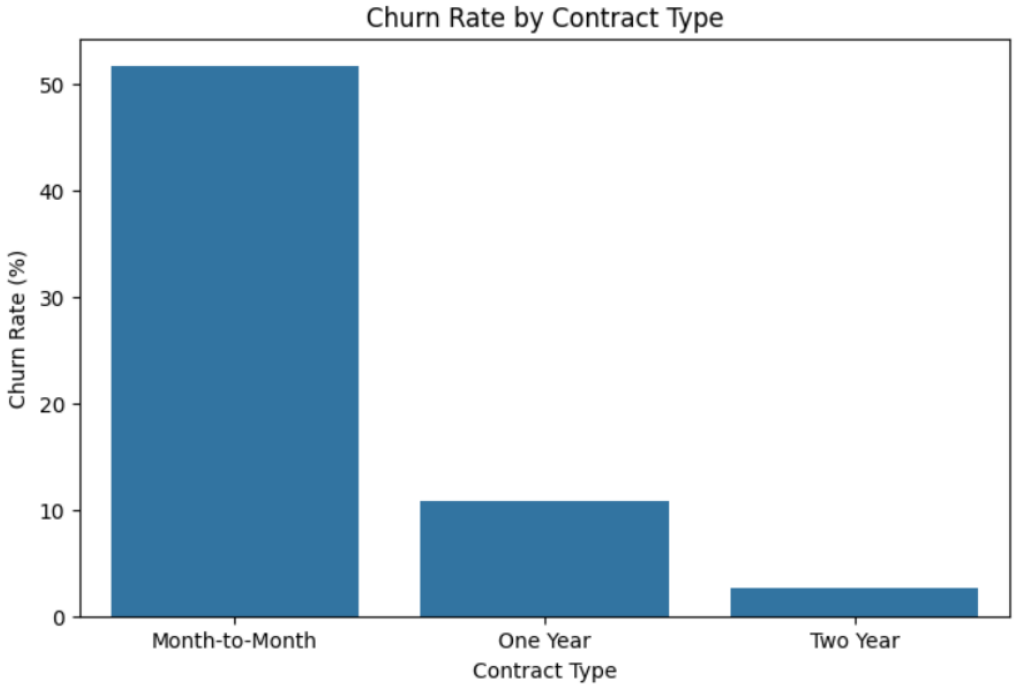
Objective	To compare customer tenure between stayed and churned customers
Interpretation	Churned customers have a lower median tenure. Stayed customers generally have longer tenure. New customers are more likely to leave.
Key Insight	Customers with shorter tenure show a higher risk of churn.
Business Insight	The company should introduce onboarding support, welcome offers and regular follow-ups during the first 12 months.

Visualization 5: Internet Type vs Churn Rate



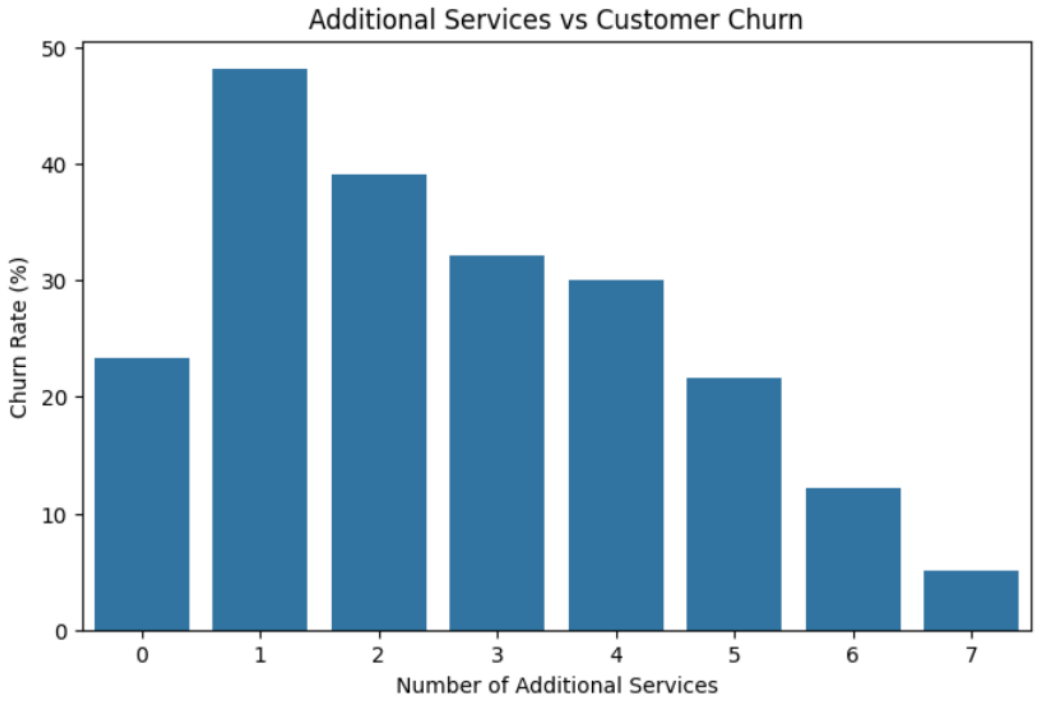
Objective	To analyze how internet service types are associated with customer churn.
Interpretation	Fiber Optic customers have the highest churn rate at 42.13%. Cable customers have a churn rate of 27.52%. DSL customers have a churn rate of 19.97%. Customers without internet service have a churn rate of 8.41%.
Key Insight	Fiber Optic customers show the highest churn rate among the internet service categories.
Business Insight	The company should investigate Fiber Optic service quality, pricing, speed and customer satisfaction to reduce churn.

Visualization 6: Contract Type vs Churn Rate



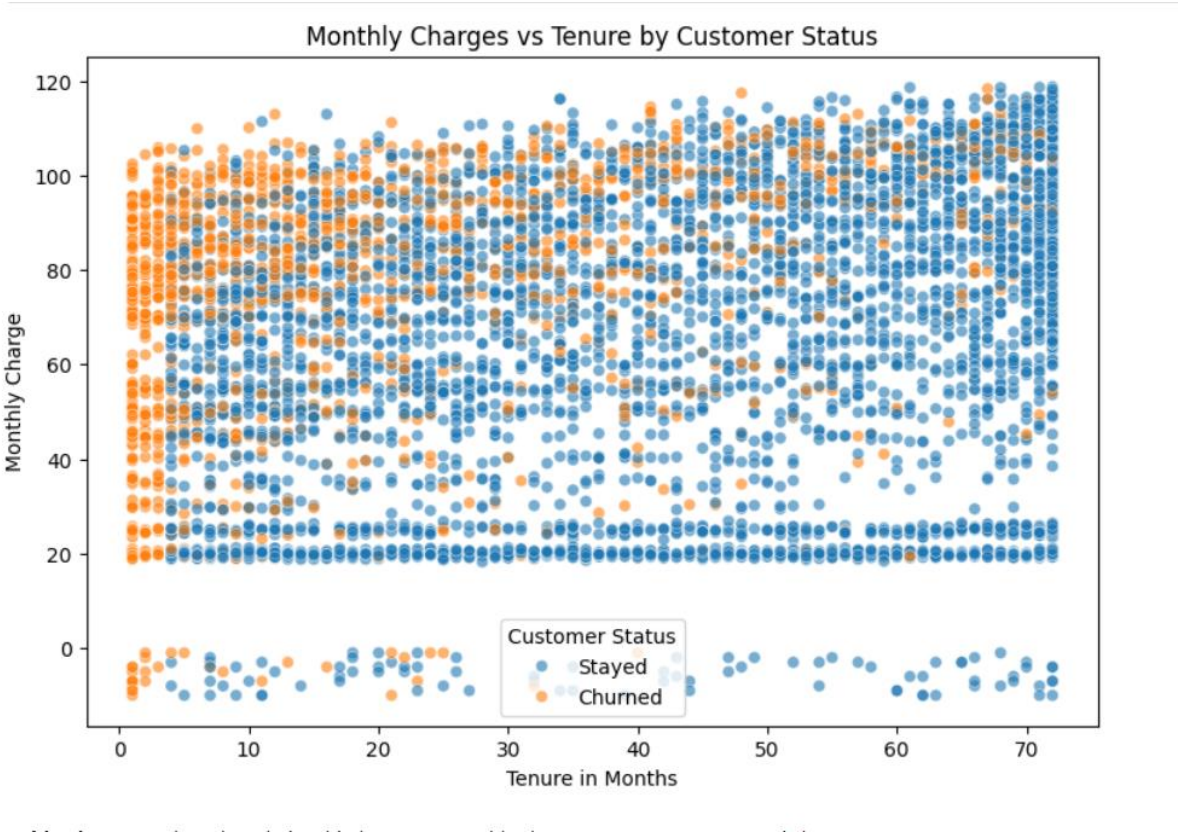
Objective	To understand the relationship between contract types and customer churn.
Interpretation	Month-to-Month customers have a 51.69% churn rate. One Year contract customers have a 10.88% churn rate. Two Year contract customers have a 2.58% churn rate.
Key Insight	Month-to-Month customers have the highest churn rate, while Two Year contract customers have the lowest.
Business Insight	The company should encourage customers to choose long-term contracts by providing attractive discounts and loyalty benefits

Visualization 7: Additional Services vs Customer Churn



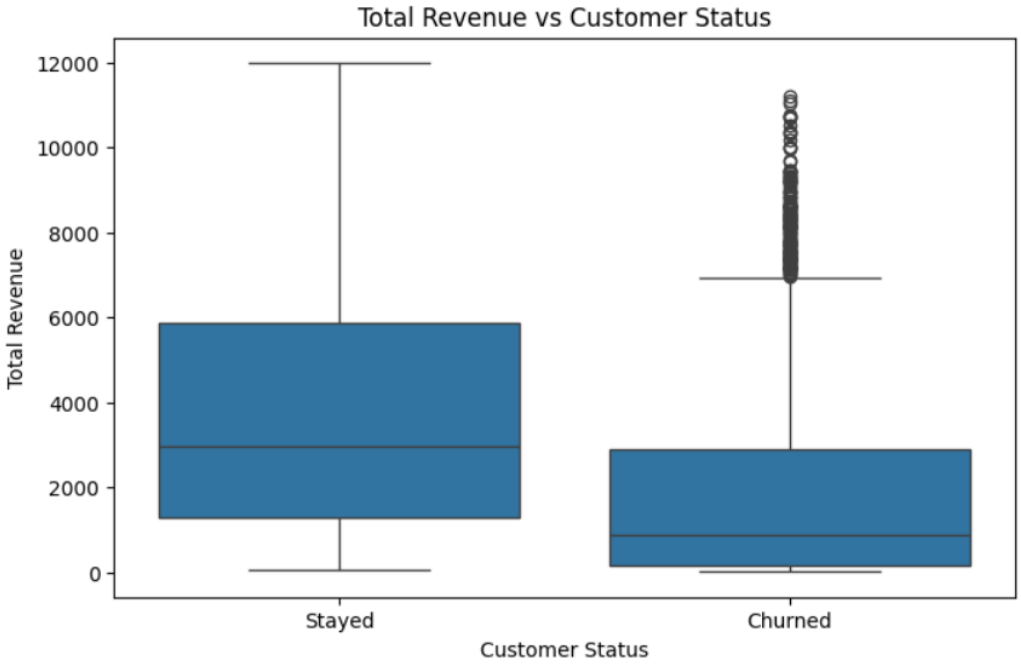
Objective	To understand whether additional services influence customer retention.
Interpretation	Customers using one additional service have a 48.11% churn rate. Customers using five additional services have a 21.62% churn rate. Customers using seven additional services have a 5.06% churn rate.
Key Insight	Customers using more additional services generally show lower churn rates.
Business Insight	The company should promote useful additional services such as online security, online backup and device protection to improve customer engagement.

Visualization 8: Monthly Charges vs Tenure by Customer Status



Objective	To analyze the relationship between monthly charges, customer tenure and churn status.
Interpretation	The chart compares monthly charges and tenure. Both stayed and churned customers have different spending patterns. Churned customers generally have shorter tenure.
Key Insight	Customers with shorter tenure and relatively higher monthly charges may require additional attention.
Business Insight	The company should review pricing plans and offer suitable packages to customers who may be at risk of leaving.

Visualization 9: Total Revenue vs Customer Status



Objective	To compare the revenue contribution of stayed and churned customers.
Interpretation	Stayed customers generate an average total revenue of 3,735.68. Churned customers generate an average total revenue of 1,971.35. Stayed customers contribute higher average revenue.
Key Insight	Retained customers generate higher average total revenue than churned customers.
Business Insight	The company should focus on customer retention strategies to maintain revenue and reduce potential revenue loss.

Summary of Key Findings

Overall churn rate among existing customers: 28.37% (1,869 churned customers out of 6,589 existing customers).

Churn by Customer Segment

Segment	Churn rate
Married: No	36.69%
Married: Yes	20.16%
Age: Above 60	37.90%
Age: 26–40	25.99%
Female	28.65%
Male	28.08%

Churn by Tenure

Tenure	Churn rate
0–12 months	59.87%
13–24 months	28.71%
25–48 months	20.39%
49–72 months	9.51%

Churn by Contract Type

Contract	Churn rate
Month-to-Month	51.69%
One Year	10.88%
Two Year	2.58%

Churn by Internet Type

Internet type	Churn rate
Fiber Optic	42.13%
Cable	27.52%
DSL	19.97%
No Internet Service	8.41%

Average Revenue Comparison

Status	Avg monthly charge	Avg total revenue
Churned	73.35	1,971.35
Stayed	61.74	3,735.68

Top Churn Reasons

Reason	Customers
Competitor had better devices	313
Competitor made better offer	311
Attitude of support person	220
Don't know	130
Competitor offered more data	117

Overall, customer tenure, contract type, internet service and competitor offers are important factors associated with churn.

8. Recommendations and Decision Support

Key finding	Recommendation
High churn among new customers	Provide welcome offers and onboarding support during the first 12 months.
High churn among Month-to-Month customers	Encourage long-term contracts through attractive benefits.
High Fiber Optic churn	Improve service quality and review pricing, speed and satisfaction.
Competitors provide better offers and devices	Introduce competitive pricing, device upgrades and better service packages.
Customer support issues	Improve customer service quality and response time.
Lower churn with more additional services	Promote useful additional services based on customer needs.
Higher revenue from stayed customers	Focus on retaining valuable existing customers.

Decision support: the company can use these findings to identify customer groups with higher churn, improve service quality, develop targeted retention strategies and protect revenue.

9. Limitations

- The findings show association, not causation. For example, customers on longer contracts may simply be those who already intended to stay.
- 120 records with negative Monthly Charge values were retained and should be investigated before pricing decisions are made.
- Joined customers (454) were excluded from churn-rate calculations, so the 28.37% rate applies to existing customers only.

10. Future Enhancements

- Develop an interactive Power BI dashboard to monitor churn, revenue and retention trends.
- Integrate customer data using SQL.
- Build a machine learning model (Logistic Regression, Decision Tree or Random Forest) to predict customer churn.
- Identify customers at high risk of leaving.
- Develop advanced customer segmentation based on spending, tenure, service usage and churn risk.
- Automate regular customer churn reporting.
- Evaluate the performance of retention campaigns over time.

11. Project Conclusion

This project identified important factors associated with telecom customer churn using Python. New customers, Month-to-Month contract customers and Fiber Optic customers have relatively higher churn rates. Competitor offers, device quality and customer support are also important areas requiring attention.

These findings can help telecom companies improve customer satisfaction, strengthen customer loyalty and reduce revenue loss.

12. Skills Demonstrated

- Data Cleaning
- Data Transformation
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Data Visualization
- Business Insights
- Data-Driven Decision Making